

```

In [24]: import pandas as pd
import numpy as np
np.random.seed(42)
df=pd.DataFrame({
    'Patient_ID':range(1,101),
    'Age':np.random.randint(35,80,100),
    'Tumor_size':np.random.uniform(1,8,100).round(2),
    'TNM_stage':np.random.choice(['I','II','III'],100),
    'ER':np.random.choice(['-','+'],100),
    'PR':np.random.choice(['-','+'],100),
    'HER2':np.random.choice(['-','+'],100),
    'Ki67':np.random.randint(5,80,100),
    'Survival_months':np.random.randint(10,120,100),
    'Status':np.random.choice([0,1],100)
})
print('数据结构: ')
print(df)
print(df.info())
print(df.describe())
df.to_csv('breast_cancer_survival.csv')
print('breast_cancer_survival.csv保存成功')
print('-----')
print('总生存曲线:')
from lifelines import KaplanMeierFitter
import matplotlib.pyplot as plt
kmf=KaplanMeierFitter()
kmf.fit(
    df['Survival_months'],
    df['Status']
)
kmf.plot()
plt.title('Overall survival curve')
plt.xlabel('Time(months)')
plt.ylabel('Survival probability')
plt.savefig(
    'Overall_survival_curve.png',
    dpi=300,
    bbox_inches='tight'
)
plt.show()
print('-----')
print('TNM分期KM曲线:')
stages=['I','II','III']
for stage in stages:
    stage_data=df[df['TNM_stage']==stage]
    kmf.fit(
        stage_data['Survival_months'],
        stage_data['Status'],
        label='Stage'+stage
    )
    kmf.plot()
plt.title('Survival curves by stage')
plt.xlabel('Time(months)')
plt.ylabel('Survival probability')
plt.legend()
plt.savefig(
    'Survival curves by stage.png',
    dpi=300,

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        bbox_inches='tight'
    )
plt.show()
print('TNM分期KM曲线的Log-rank分析: ')
from lifelines.statistics import multivariate_logrank_test
result_stage=multivariate_logrank_test(
    df['Survival_months'],
    df['Status'],
    df['TNM_stage']
)
print('TNM分期的KM曲线p值: ',result_stage.p_value)
print('因为p值>0.05,所以TNM分期的生存曲线没有统计学意义。')
print('-----')
print('ER的KM曲线: ')
for group_er in ['-','+']:
    ix=df['ER']==group_er
    kmf.fit(
        df.loc[ix,'Survival_months'],
        df.loc[ix,'Status'],
        label='ER'+group_er
    )
    kmf.plot()
plt.title('Survival curves by ER')
plt.xlabel('Time(months)')
plt.ylabel('Survival probability')
plt.legend()
plt.savefig(
    'Survival curves by ER.png',
    dpi=300,
    bbox_inches='tight'
)
plt.show()
print('ER的KM曲线的Log-rank检验: ')
from lifelines.statistics import logrank_test
result_ER=logrank_test(
    df['Survival_months'],
    df['Status'],
    df['ER']
)
print('ER的KM曲线p值: ',result_ER.p_value)
print('因为P值<0.05,所以ER的KM曲线有统计学意义。')
print('-----')
print('PR的KM曲线: ')
for group_pr in ['-','+']:
    ix=df['PR']==group_pr
    kmf.fit(
        df.loc[ix,'Survival_months'],
        df.loc[ix,'Status'],
        label='PR'+group_pr
    )
    kmf.plot()
plt.title('Survival curves by PR')
plt.xlabel('Time(months)')
plt.ylabel('Survival probability')
plt.legend()
plt.savefig(
    'Survival curves by PR.png',
    dpi=300,
    bbox_inches='tight'
)
)

```

```

plt.show()
print('PR的KM曲线的Log-rank检验: ')
result_PR=logrank_test(
    df['Survival_months'],
    df['Status'],
    df['PR']
)
print('PR的KM曲线P值: ',result_PR.p_value)
print('因为p值<0.05,所以PR的KM曲线有统计学意义。')
print('-----')
print('HER2的KM曲线: ')
for group_her2 in ['-','+']:
    ix=df['HER2']==group_her2
    kmf.fit(
        df.loc[ix,'Survival_months'],
        df.loc[ix,'Status'],
        label='HER2'+group_her2
    )
    kmf.plot()
plt.title('Survival curves by HER2')
plt.xlabel('Time(months)')
plt.ylabel('Survival probability')
plt.legend()
plt.savefig(
    'Survival curves by HER2.png',
    dpi=300,
    bbox_inches='tight'
)
plt.show()
print('HER2的KM曲线的Log-rank检验: ')
result_HER2=logrank_test(
    df['Survival_months'],
    df['Status'],
    df['HER2']
)
print('HER2的KM曲线p值: ',result_HER2.p_value)
print('因为p<0.05,所以HER2的KM曲线有统计学意义。')
print('-----')
print('Ki67的KM曲线: ')
df['Ki67_group']=df['Ki67'].apply(lambda x:'high' if x>=30 else 'low')
for group_ki67 in ['low','high']:
    ix=df['Ki67_group']==group_ki67
    kmf.fit(
        df.loc[ix,'Survival_months'],
        df.loc[ix,'Status'],
        label='Ki67'+group_ki67
    )
    kmf.plot()
plt.title('Survival curves by Ki67')
plt.xlabel('Time(months)')
plt.ylabel('Survival probability')
plt.legend()
plt.savefig(
    'Survival curves by Ki67.png',
    dpi=300,
    bbox_inches='tight'
)
plt.show()
print('Ki67的KM曲线Log-rank检验: ')
result_Ki67=logrank_test(

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    df['Survival_months'],
    df['Status'],
    df['Ki67_group']
)
print('Ki67的KM曲线p值: ', result_Ki67.p_value)
print('因为p值<0.05, 所以Ki67的KM曲线有统计学意义。')
print('-----')
print('单因素Cox回归: ')
from lifelines import CoxPHFitter
cph=CoxPHFitter()
cph.fit(
    df[['Survival_months', 'Status', 'Age']],
    duration_col='Survival_months',
    event_col='Status'
)
print('Age:')
cph.print_summary()
print('C-index:', cph.concordance_index_)
print('因为C-index=0.49, 故Age的Cox回归模型属于完全随机预测。且p>0.05, ER对生存结局')
print('-----')
print('ER:')
df['ER_binary']=(df['ER']=='+').astype(int)
cph.fit(
    df[['Survival_months', 'Status', 'ER_binary']],
    duration_col='Survival_months',
    event_col='Status'
)
cph.print_summary()
print('C-index:', cph.concordance_index_)

print('-----')
print('PR:')
df['PR_binary']=(df['PR']=='+').astype(int)
cph.fit(
    df[['Survival_months', 'Status', 'PR_binary']],
    duration_col='Survival_months',
    event_col='Status'
)
cph.print_summary()
print('C-index:', cph.concordance_index_)
print('因为C-index=0.48, 故PR的Cox回归模型属于完全随机预测。且p>0.05, PR对生存结局')
print('-----')
print('TNM_stage:')
df['Stage_binary']=(df['TNM_stage']=='III').astype(int)
cph.fit(
    df[['Survival_months', 'Status', 'Stage_binary']],
    duration_col='Survival_months',
    event_col='Status'
)
cph.print_summary()
print('C-index:', cph.concordance_index_)
print('因为C-index=0.56, 故TNM-stage的Cox回归模型属于完全随机预测。且p>0.05, TNM分')
print('-----')
print('Tumor_size:')
cph.fit(
    df[['Survival_months', 'Status', 'Tumor_size']],
    duration_col='Survival_months',
    event_col='Status'
)
cph.print_summary()

```

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print('C-index:', cph.concordance_index_)
print('因为C-index=0.48, 故Tumor_size的Cox回归模型属于完全随机预测。且 $p > 0.05$ , 肿瘤
print('-----')
print('HER2:')
df['HER2_binary']=(df['HER2']=='+').astype(int)
cph.fit(
    df[['Survival_months', 'Status', 'HER2_binary']],
    duration_col='Survival_months',
    event_col='Status'
)
cph.print_summary()
print('C-index:', cph.concordance_index_)
print('因为C-index=0.49, 故HER2的Cox回归模型属于完全随机预测。且 $p > 0.05$ , HER2对生存
print('-----')
print('Ki67:')
df['Ki67_binary']=(df['Ki67']>=30).astype(int)
cph.fit(
    df[['Survival_months', 'Status', 'Ki67_binary']],
    duration_col='Survival_months',
    event_col='Status'
)
cph.print_summary()
print('C-index:', cph.concordance_index_)
print('因为C-index=0.53, 故Ki67的Cox回归模型属于完全随机预测。且 $p > 0.05$ , Ki67对生存
print('-----')
print('因为所有可能影响生存的变量的P值均大于0.05, 所以这些都不是影响生存结局的相关
print('假设Age、Tumor_size、TNM_stage是单因素显著变量, 进行下一步的多因素Cox分析,
print('-----')
print('多因素回归分析: ')
cph.fit(
    df[['Survival_months', 'Status', 'Age', 'Tumor_size', 'Stage_binary']],
    duration_col='Survival_months',
    event_col='Status'
)
cph.print_summary()
print('因为Age、Tumor_size、TNM_stage的P值均大于0.05, 所以均没有统计学意义。')
print('因为C-index=0.59, 所以Age、Tumor_size、TNM_stage的多因素Cox回归分析模型属于

```

数据结构:

	Patient_ID	Age	Tumor_size	TNM_stage	ER	PR	HER2	Ki67	Survival_months	\
0	1	73	5.98	III	+	+	-	34		24
1	2	63	6.53	I	-	-	+	21		40
2	3	49	5.24	I	-	+	+	66		10
3	4	77	7.48	III	-	+	-	17		63
4	5	42	5.56	II	+	+	+	63		12
..	...	...	...	...	..	..	...	...		...
95	96	78	3.28	II	+	+	-	61		87
96	97	42	2.54	I	+	+	-	22		82
97	98	58	5.98	III	-	-	-	51		119
98	99	45	6.67	III	-	+	-	53		10
99	100	51	3.44	I	+	+	-	18		60

	Status
0	0
1	0
2	1
3	1
4	0
..	...
95	0
96	1
97	0
98	0
99	1

[100 rows x 10 columns]

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 100 entries, 0 to 99

Data columns (total 10 columns):

#	Column	Non-Null Count	Dtype
0	Patient_ID	100 non-null	int64
1	Age	100 non-null	int32
2	Tumor_size	100 non-null	float64
3	TNM_stage	100 non-null	object
4	ER	100 non-null	object
5	PR	100 non-null	object
6	HER2	100 non-null	object
7	Ki67	100 non-null	int32
8	Survival_months	100 non-null	int32
9	Status	100 non-null	int64

dtypes: float64(1), int32(3), int64(2), object(4)

memory usage: 6.8+ KB

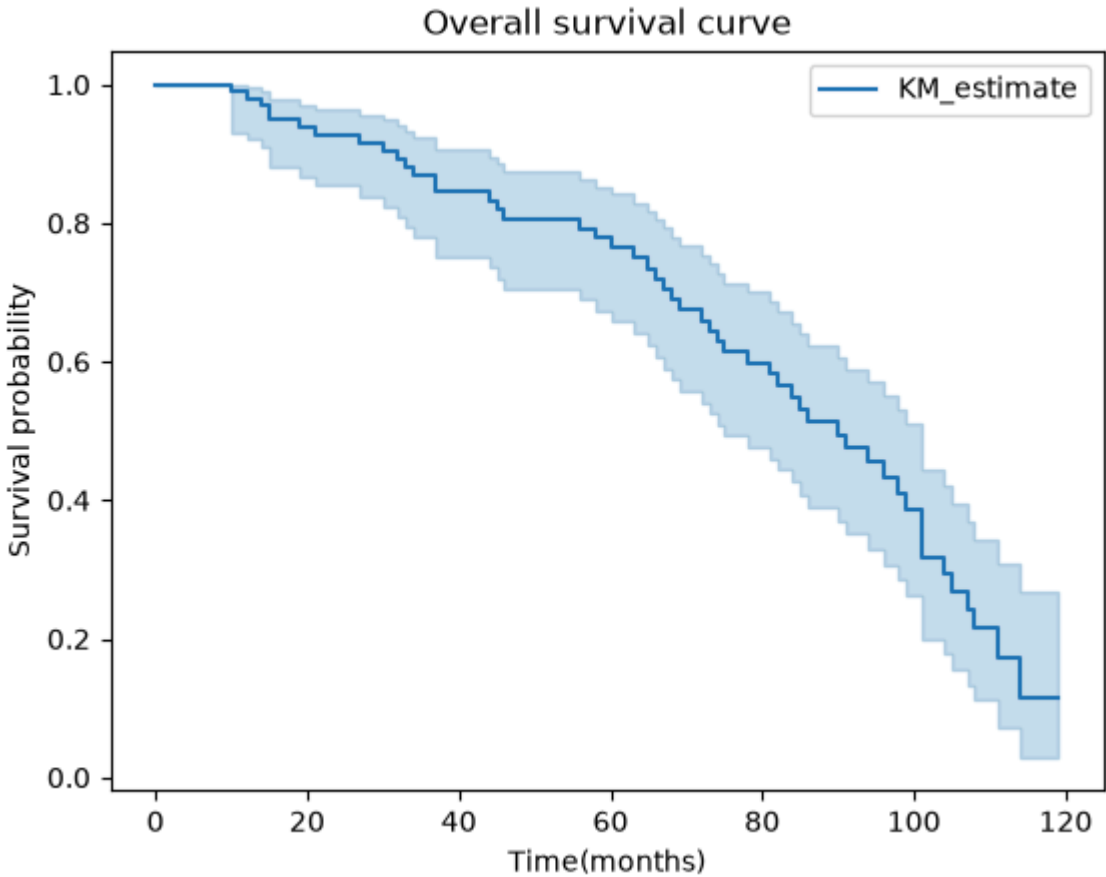
None

	Patient_ID	Age	Tumor_size	Ki67	Survival_months	\
count	100.000000	100.000000	100.000000	100.000000	100.000000	
mean	50.500000	56.600000	4.464400	41.710000	62.72000	
std	29.011492	13.311346	1.883054	21.216201	32.60417	
min	1.000000	35.000000	1.000000	5.000000	10.00000	
25%	25.750000	45.000000	2.850000	22.750000	34.75000	
50%	50.500000	56.500000	4.360000	43.000000	64.50000	
75%	75.250000	68.250000	5.870000	61.000000	91.25000	
max	100.000000	79.000000	7.980000	79.000000	119.00000	

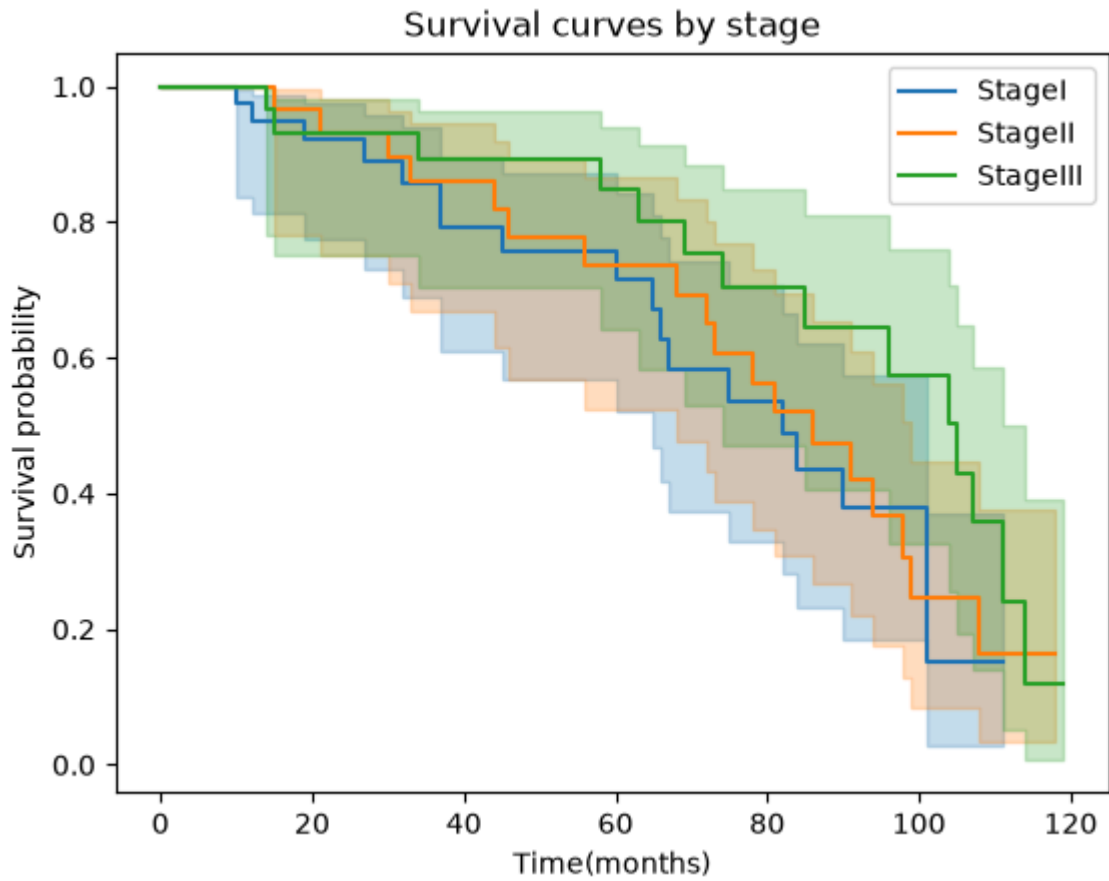
	Status
count	100.000000
mean	0.510000
std	0.502418

min 0.000000
25% 0.000000
50% 1.000000
75% 1.000000
max 1.000000
breast_cancer_survival.csv保存成功

总生存曲线：



TNM分期KM曲线：

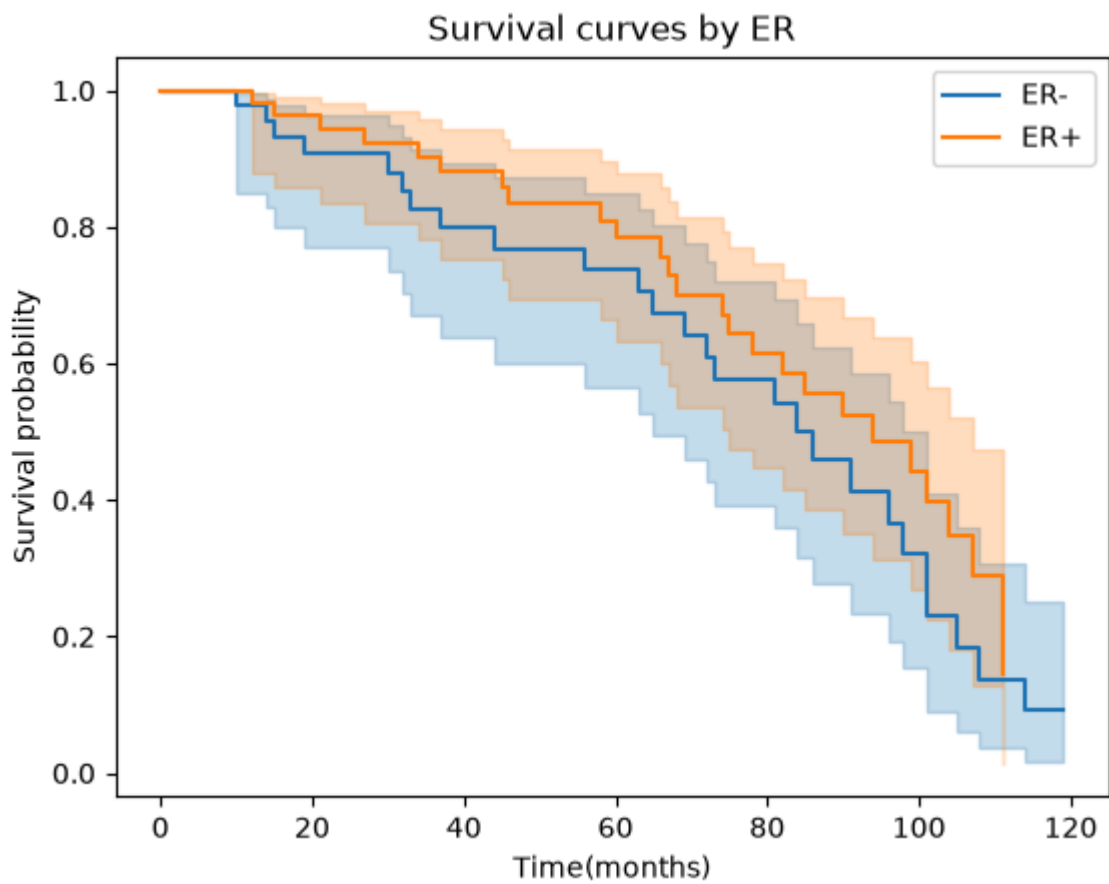


TNM分期KM曲线的Log-rank分析：

TNM分期的KM曲线p值： 0.9662874186389613

因为p值>0.05,所以TNM分期的生存曲线没有统计学意义。

ER的KM曲线：

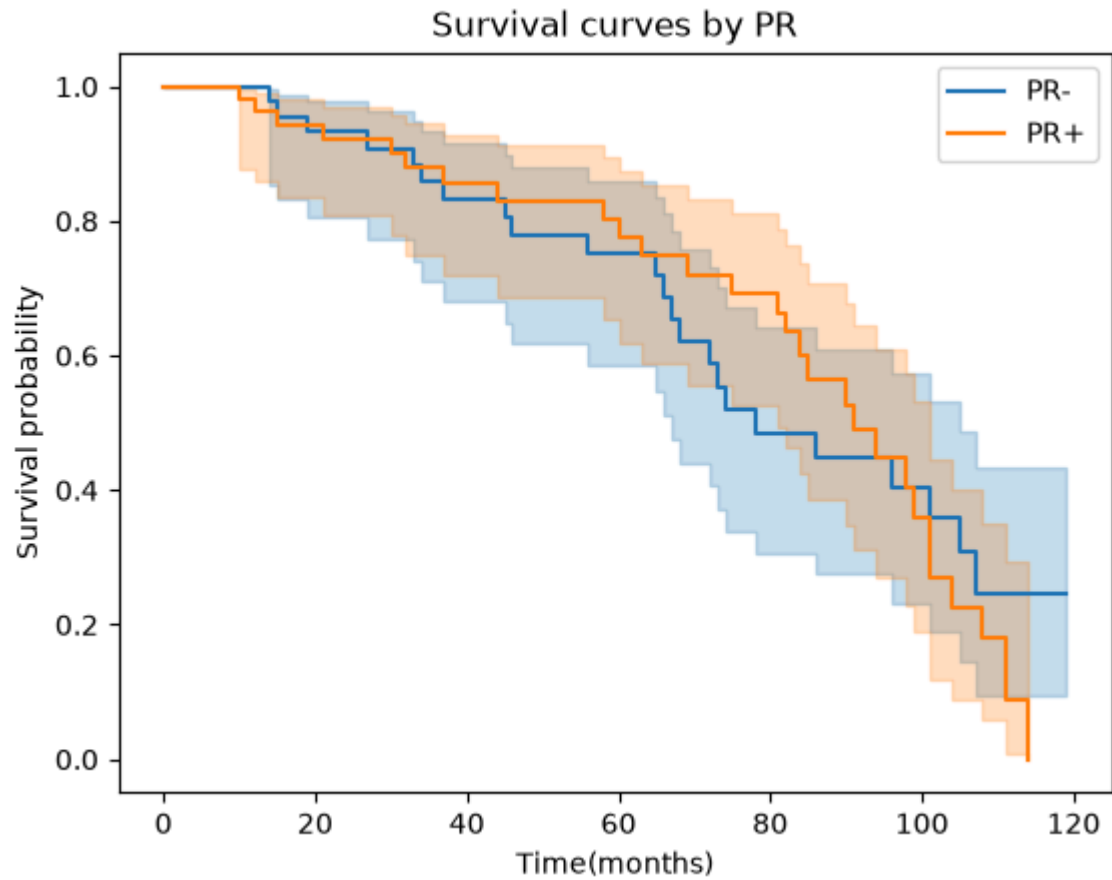


ER的KM曲线的Log-rank检验:

ER的KM曲线p值: 1.3007720145188643e-45

因为P值<0.05, 所以ER的KM曲线有统计学意义。

PR的KM曲线:

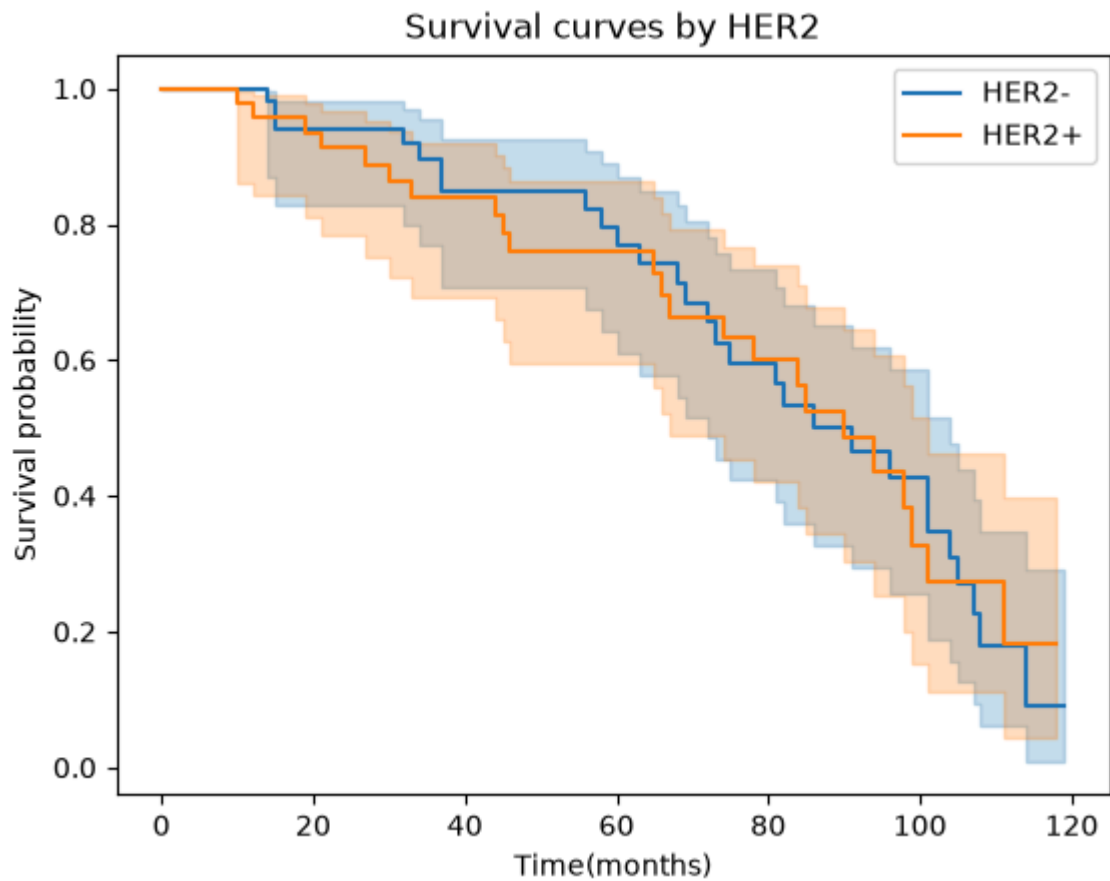


PR的KM曲线的Log-rank检验:

PR的KM曲线P值: 1.3007720145188643e-45

因为p值<0.05, 所以PR的KM曲线有统计学意义。

HER2的KM曲线:

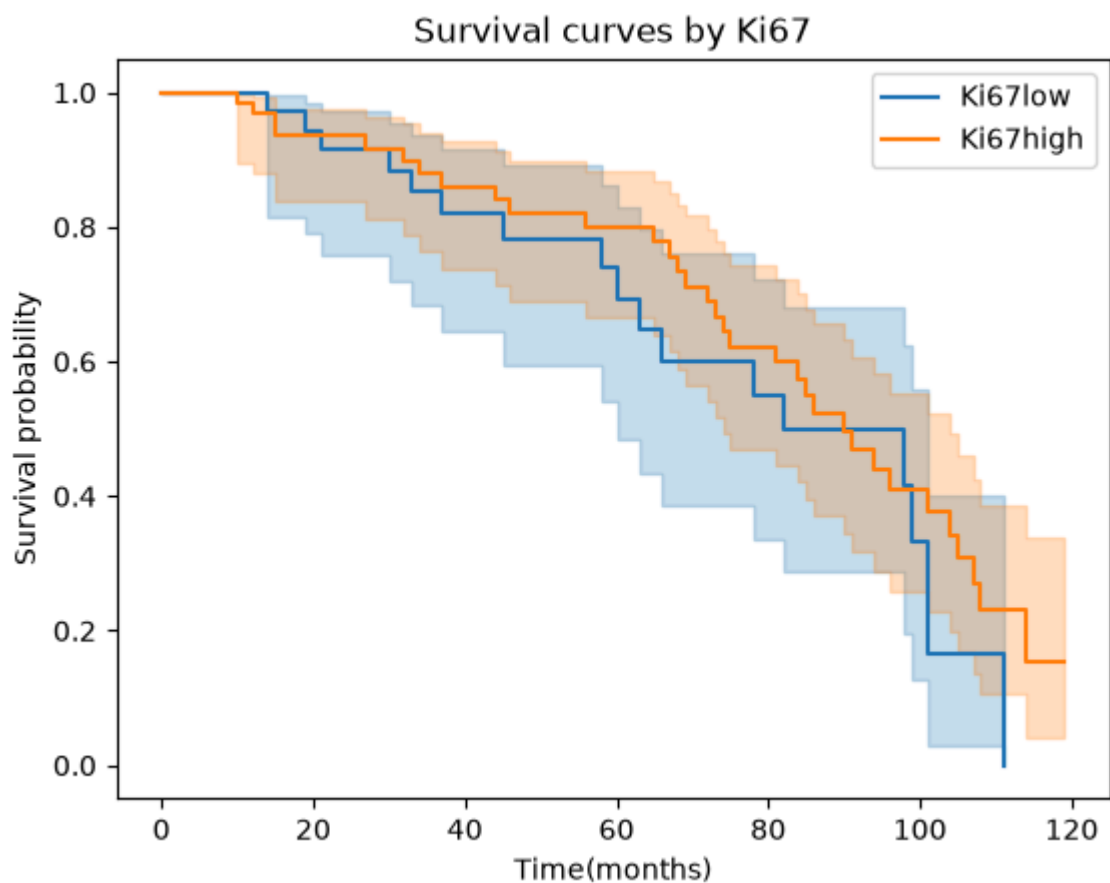


HER2的KM曲线的Log-rank检验:

HER2的KM曲线p值: 1.3007720145188643e-45

因为 $p < 0.05$, 所以HER2的KM曲线有统计学意义。

Ki67的KM曲线:



Ki67的KM曲线Log-rank检验:
Ki67的KM曲线p值: 1.3007720145188643e-45
因为p值<0.05, 所以Ki67的KM曲线有统计学意义。

单因素Cox回归:
Age:

```
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <tbody> <th>model</th> <th>duration col</th> <th>event col</th>
<th>baseline estimation</th> <th>number of observations</th> <th>number of
events observed</th> <th>partial log-likelihood</th> <th>time fit was run</th>
</tbody> <tbody> </tbody>
```

lifelines.CoxPHFitter

'Survival_months'
'Status'
breslow
100
51
-185.29
2026-07-27 13:08:11 UTC

```
<thead> <th style="min-width: 12px;"></th> <th style="min-width:
12px;">coef</th> <th style="min-width: 12px;">exp(coef)</th> <th style="min-
width: 12px;">se(coef)</th> <th style="min-width: 12px;">coef lower 95%</th>
<th style="min-width: 12px;">coef upper 95%</th> <th style="min-width:
12px;">exp(coef) lower 95%</th> <th style="min-width: 12px;">exp(coef) upper
95%</th> <th style="min-width: 12px;">cmp to</th> <th style="min-width:
12px;">z</th> <th style="min-width: 12px;">p</th> <th style="min-width:
12px;">-log2(p)</th> </thead> <tbody> <th>Age</th> </tbody> <tbody>
</tbody>
```

0.00	1.00	0.01	-0.02	0.02	0.98	1.02	0.00	0.01	0.99	0.01
------	------	------	-------	------	------	------	------	------	------	------

```
<br>
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <tbody> <th>Concordance</th> <th>Partial AIC</th> <th>log-
likelihood ratio test</th> <th>-log2(p) of ll-ratio test</th> </tbody> <tbody>
</tbody>
```

0.49
372.59
0.00 on 1 df
0.01

C-index: 0.4899791231732777

因为C-index=0.49，故Age的Cox回归模型属于完全随机预测。且 $p>0.05$ ，ER对生存结局之间没有统计学关联。

ER:

```
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
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<th>baseline estimation</th> <th>number of observations</th> <th>number of
events observed</th> <th>partial log-likelihood</th> <th>time fit was run</th>
</tbody> <tbody> </tbody>
```

lifelines.CoxPHFitter

'Survival_months'
'Status'
breslow
100
51
-184.69
2026-07-27 13:08:11 UTC

```
<thead> <th style="min-width: 12px;"></th> <th style="min-width:
12px;">coef</th> <th style="min-width: 12px;">exp(coef)</th> <th style="min-
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95%</th> <th style="min-width: 12px;">cmp to</th> <th style="min-width:
12px;">z</th> <th style="min-width: 12px;">p</th> <th style="min-width:
12px;">-log2(p)</th> </thead> <tbody> <th>ER_binary</th> </tbody> <tbody>
</tbody>
```

-0.31	0.73	0.28	-0.87	0.24	0.42	1.28	0.00	-1.10	0.27	1.88
-------	------	------	-------	------	------	------	------	-------	------	------


```
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
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</style> <tbody> <th>Concordance</th> <th>Partial AIC</th> <th>log-
likelihood ratio test</th> <th>-log2(p) of ll-ratio test</th> </tbody> <tbody>
</tbody>
```

0.55
371.38
1.20 on 1 df
1.87

C-index: 0.5457202505219206

PR:

```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <tbody> <th>model</th> <th>duration col</th> <th>event col</th>
<th>baseline estimation</th> <th>number of observations</th> <th>number of
events observed</th> <th>partial log-likelihood</th> <th>time fit was run</th>
</tbody> <tbody> </tbody>

```

lifelines.CoxPHFitter

'Survival_months'

'Status'

breslow

100

51

-185.26

2026-07-27 13:08:11 UTC

```

<thead> <th style="min-width: 12px;"></th> <th style="min-width:
12px;">coef</th> <th style="min-width: 12px;">exp(coef)</th> <th style="min-
width: 12px;">se(coef)</th> <th style="min-width: 12px;">coef lower 95%</th>
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12px;">exp(coef) lower 95%</th> <th style="min-width: 12px;">exp(coef) upper
95%</th> <th style="min-width: 12px;">cmp to</th> <th style="min-width:
12px;">z</th> <th style="min-width: 12px;">p</th> <th style="min-width:
12px;">-log2(p)</th> </thead> <tbody> <th>PR_binary</th> </tbody> <tbody>
</tbody>

```

0.08 1.08 0.28 -0.48 0.63 0.62 1.88 0.00 0.27 0.79 0.34


```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
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</style> <tbody> <th>Concordance</th> <th>Partial AIC</th> <th>log-
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</tbody>

```

0.48

372.51

0.07 on 1 df

0.34

C-index: 0.4812108559498956

因为C-index=0.48，故PR的Cox回归模型属于完全随机预测。且 $p>0.05$ ，PR对生存结局之间没有统计学关联。

TNM_stage:

```
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
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</style> <tbody> <th>model</th> <th>duration col</th> <th>event col</th>
<th>baseline estimation</th> <th>number of observations</th> <th>number of
events observed</th> <th>partial log-likelihood</th> <th>time fit was run</th>
</tbody> <tbody> </tbody>
```

lifelines.CoxPHFitter

'Survival_months'

'Status'

breslow

100

51

-183.83

2026-07-27 13:08:11 UTC

```
<thead> <th style="min-width: 12px;"></th> <th style="min-width:
12px;">coef</th> <th style="min-width: 12px;">exp(coef)</th> <th style="min-
width: 12px;">se(coef)</th> <th style="min-width: 12px;">coef lower 95%</th>
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12px;">exp(coef) lower 95%</th> <th style="min-width: 12px;">exp(coef) upper
95%</th> <th style="min-width: 12px;">cmp to</th> <th style="min-width:
12px;">z</th> <th style="min-width: 12px;">p</th> <th style="min-width:
12px;">-log2(p)</th> </thead> <tbody> <th>Stage_binary</th> </tbody>
<tbody> </tbody>
```

-0.53 0.59 0.32 -1.15 0.10 0.32 1.10 0.00 -1.65 0.10 3.35


```
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <tbody> <th>Concordance</th> <th>Partial AIC</th> <th>log-
likelihood ratio test</th> <th>-log2(p) of ll-ratio test</th> </tbody> <tbody>
</tbody>
```

0.56

369.66

2.92 on 1 df

3.52

C-index: 0.5588726513569937

因为C-index=0.56，故TNM-stage的Cox回归模型属于完全随机预测。且 $p>0.05$ ，TNM分期对生存结局之间没有统计学关联。

Tumor_size:

```
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <tbody> <th>model</th> <th>duration col</th> <th>event col</th>
<th>baseline estimation</th> <th>number of observations</th> <th>number of
events observed</th> <th>partial log-likelihood</th> <th>time fit was run</th>
</tbody> <tbody> </tbody>
```

lifelines.CoxPHFitter

'Survival_months'
'Status'
breslow
100
51
-185.23
2026-07-27 13:08:11 UTC

```
<thead> <th style="min-width: 12px;"></th> <th style="min-width:
12px;">coef</th> <th style="min-width: 12px;">exp(coef)</th> <th style="min-
width: 12px;">se(coef)</th> <th style="min-width: 12px;">coef lower 95%</th>
<th style="min-width: 12px;">coef upper 95%</th> <th style="min-width:
12px;">exp(coef) lower 95%</th> <th style="min-width: 12px;">exp(coef) upper
95%</th> <th style="min-width: 12px;">cmp to</th> <th style="min-width:
12px;">z</th> <th style="min-width: 12px;">p</th> <th style="min-width:
12px;">-log2(p)</th> </thead> <tbody> <th>Tumor_size</th> </tbody>
<tbody> </tbody>
```

-0.03	0.97	0.08	-0.18	0.12	0.83	1.13	0.00	-0.36	0.72	0.48
-------	------	------	-------	------	------	------	------	-------	------	------


```
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <tbody> <th>Concordance</th> <th>Partial AIC</th> <th>log-
likelihood ratio test</th> <th>-log2(p) of ll-ratio test</th> </tbody> <tbody>
</tbody>
```

0.48

372.45

0.13 on 1 df

0.48

C-index: 0.4793319415448852

因为C-index=0.48，故Tumor_size的Cox回归模型属于完全随机预测。且 $p>0.05$ ，肿瘤大小对生存结局之间没有统计学关联。

HER2:

```
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <tbody> <th>model</th> <th>duration col</th> <th>event col</th>
<th>baseline estimation</th> <th>number of observations</th> <th>number of
events observed</th> <th>partial log-likelihood</th> <th>time fit was run</th>
</tbody> <tbody> </tbody>
```

lifelines.CoxPHFitter

'Survival_months'
'Status'
breslow
100
51
-185.29
2026-07-27 13:08:12 UTC

```
<thead> <th style="min-width: 12px;"></th> <th style="min-width:
12px;">coef</th> <th style="min-width: 12px;">exp(coef)</th> <th style="min-
width: 12px;">se(coef)</th> <th style="min-width: 12px;">coef lower 95%</th>
<th style="min-width: 12px;">coef upper 95%</th> <th style="min-width:
12px;">exp(coef) lower 95%</th> <th style="min-width: 12px;">exp(coef) upper
95%</th> <th style="min-width: 12px;">cmp to</th> <th style="min-width:
12px;">z</th> <th style="min-width: 12px;">p</th> <th style="min-width:
12px;">-log2(p)</th> </thead> <tbody> <th>HER2_binary</th> </tbody>
<tbody> </tbody>
```

-0.01	0.99	0.28	-0.57	0.54	0.57	1.72	0.00	-0.04	0.97	0.05
-------	------	------	-------	------	------	------	------	-------	------	------

```
<br>
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
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</style> <tbody> <th>Concordance</th> <th>Partial AIC</th> <th>log-
likelihood ratio test</th> <th>-log2(p) of ll-ratio test</th> </tbody> <tbody>
</tbody>
```

0.49
372.58
0.00 on 1 df
0.05

C-index: 0.48851774530271397
因为C-index=0.49，故HER2的Cox回归模型属于完全随机预测。且p>0.05,HER2对生存结局之间没有统计学关联。

Ki67:

```
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <tbody> <th>model</th> <th>duration col</th> <th>event col</th>
<th>baseline estimation</th> <th>number of observations</th> <th>number of
events observed</th> <th>partial log-likelihood</th> <th>time fit was run</th>
</tbody> <tbody> </tbody>
```

lifelines.CoxPHFitter

'Survival_months'

'Status'

breslow

100

51

-184.86

2026-07-27 13:08:12 UTC

```
<thead> <th style="min-width: 12px;"></th> <th style="min-width:
12px;">coef</th> <th style="min-width: 12px;">exp(coef)</th> <th style="min-
width: 12px;">se(coef)</th> <th style="min-width: 12px;">coef lower 95%</th>
<th style="min-width: 12px;">coef upper 95%</th> <th style="min-width:
12px;">exp(coef) lower 95%</th> <th style="min-width: 12px;">exp(coef) upper
95%</th> <th style="min-width: 12px;">cmp to</th> <th style="min-width:
12px;">z</th> <th style="min-width: 12px;">p</th> <th style="min-width:
12px;">-log2(p)</th> </thead> <tbody> <th>Ki67_binary</th> </tbody>
<tbody> </tbody>
```

-0.28 0.75 0.30 -0.86 0.30 0.42 1.35 0.00 -0.95 0.34 1.54


```
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
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</style> <tbody> <th>Concordance</th> <th>Partial AIC</th> <th>log-
likelihood ratio test</th> <th>-log2(p) of ll-ratio test</th> </tbody> <tbody>
</tbody>
```

0.53

371.72

0.87 on 1 df

1.51

C-index: 0.5258872651356994

因为C-index=0.53, 故Ki67的Cox回归模型属于完全随机预测。且 $p>0.05$, Ki67对生存结局之间没有统计学关联。

因为所有可能影响生存的变量的P值均大于0.05, 所以这些都不是影响生存结局的相关变量, 无需进行下一步的多因素Cox。

假设Age、Tumor_size、TNM_stage是单因素显著变量, 进行下一步的多因素Cox分析, 以筛选出独立危险因素。

多因素回归分析:

```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
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<th>baseline estimation</th> <th>number of observations</th> <th>number of
events observed</th> <th>partial log-likelihood</th> <th>time fit was run</th>
</tbody> <tbody> </tbody>

```

lifelines.CoxPHFitter

'Survival_months'
'Status'
breslow
100
51
-183.81
2026-07-27 13:08:12 UTC

```

<thead> <th style="min-width: 12px;"></th> <th style="min-width:
12px;">coef</th> <th style="min-width: 12px;">exp(coef)</th> <th style="min-
width: 12px;">se(coef)</th> <th style="min-width: 12px;">coef lower 95%</th>
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12px;">exp(coef) lower 95%</th> <th style="min-width: 12px;">exp(coef) upper
95%</th> <th style="min-width: 12px;">cmp to</th> <th style="min-width:
12px;">z</th> <th style="min-width: 12px;">p</th> <th style="min-width:
12px;">-log2(p)</th> </thead> <tbody> <th>Age</th> <th>Tumor_size</th>
<th>Stage_binary</th> </tbody> <tbody></tbody>

```

0.00	1.00	0.01	-0.02	0.02	0.98	1.02	0.00	0.13	0.89	0.16
0.01	1.01	0.08	-0.15	0.18	0.86	1.20	0.00	0.17	0.87	0.21
-0.55	0.58	0.34	-1.20	0.11	0.30	1.12	0.00	-1.63	0.10	3.28


```

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likelihood ratio test</th> <th>-log2(p) of ll-ratio test</th> </tbody> <tbody>
</tbody>

```

0.59
373.63
2.96 on 3 df
1.33

因为Age、Tumor_size、TNM_stage的P值均大于0.05，所以均没有统计学意义。
因为C-index=0.59,所以Age、Tumor_size、TNM_stage的多因素Cox回归分析模型属于随机预测模型。

In []: